Nathaniel Pritchard
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CONTACT

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RESEARCH INTERESTS

Randomized Numerical Linear Algebra. Statistical Computation. Generalized Linear Models. Optimization. High Performance Computing.

EDUCATION

University of Wisconsin-Madison Madison, WI
Ph.D., Statistics August 2018 - August 2024
Thesis Title: Facilitating Large-Scale Logistic Regression Through Rigorously Tracked Iterative Right Random Sketching
Adviser: Vivak Patel

University of North Carolina at Chapel Hill Chapel Hill, NC
B.S., Statistics and Analytics August 2014 - May 2018
(Highest Honors and Highest Distinction)
Thesis Title: Metropolitan Community Delineation
Honors Thesis Adviser: Shankar Bhamidi

RESEARCH EMPLOYMENT

University of Oxford, Oxford, United Kingdom
Postdoctoral Research Associate October 2024 - October 2026
Supervisors: Yuji Nakatsukasa and Per-Gunnar Martinsson
Topic: Certifying accuracy of randomized algorithms in numerical linear algebra

University of Wisconsin-Madison, Madison, WI
Research Assistant August 2023 - May 2024
Supervisor: Vivak Patel
Topic: Development of Randomized Linear Algebra Library for Julia

Argonne National Laboratory, Lemont, IL
Givens Associate May 2023 - July 2023
Supervisor: Adrian Maldonado
Topic: Accelerating Newton-Raphson on GPUs using deflation methods

Argonne National Laboratory, Lemont, IL
Givens Associate May 2020 - July 2020
Supervisor: Adrian Maldonado
Topic: Preconditioners for solving graph Laplacians arising from power grid networks

PUBLICATIONS

Pritchard, N., Park, T., Nakatsukasa, Y., & Martinsson, P.-G. (2026). Fast Rank Adaptive CUR via a Recycled Small Sketch. *SIAM Journal on Scientific Computing*, to appear. arXiv:2509.21963.

Lazzarino, L., Pearce, K., & **Pritchard, N.** (2026). Efficient error estimators for Generalized Nyström. arXiv preprint arXiv:2601.11493. (Accepted to ENUMATH Conference Proceedings)

Pritchard, N., & Patel, V. (2024). Solving, Tracking and Stopping Streaming Linear Inverse Problems. *Inverse Problems*, 40 (8), 10.1088/1361-6420/ad5583.

Pritchard, N., & Patel, V. (2023). Towards Practical Large-Scale Randomized Iterative Least Squares Solvers through Uncertainty Quantification. *SIAM/ASA Journal on Uncertainty Quantification*, 11(3), 10.1137/22M1515057.

He, M., Glasser, J., **Pritchard, N.**, Bhamidi, S., & Kaza, N. (2020). Demarcating geographic regions using community detection in commuting networks with significant self-loops. *PLOS ONE* 15(4), e0230941.

PRE-PRINTS

Lazzarino, L., Pearce, K., & **Pritchard, N.** (2026). Blind error for CUR approximation. arXiv preprint arXiv:2609.23810

Patel, V., Maldonado, D. A., Melnichenko, M., **Pritchard, N.**, Rao, V., Rebrova, E., & Sankararaman, S. (2025). Scientific Applications Leveraging Randomized Linear Algebra. arXiv preprint arXiv:2506.16457.

IN PREPARATION

Pritchard, N., Pearce, Nakatsukasa, Y., Martinsson, P.-G., & K., Kileel, J., IterativeTensorCUR

Pritchard, N., Patel, V., & Maldonado, D. A., RandLinearAlgebra.jl: A Julia Library for prototyping RandNLA algorithms

Halikias D., **Pritchard, N.**, & Nakatsukasa, Y., Divide-without-conquer: an eigensolver exploiting localization

TEACHING EXPERIENCE

Teaching Assistant (Undergraduate): Introductory Applied Statistics for the Life Sciences (Fall 2018 - Spring 2019), Introduction to Statistical Methods (Fall 2019), Introduction to Statistics for Science and Engineering (Spring 2019), Data Science Modeling I (Fall 2020 - Spring 2022, Spring 2023)**
Teaching Assistant (Graduate): Statistical Methods I (Fall 2022)
Tutor (Oxford): Continuous Optimization (Spring 2026)
Instructor of Record: Data Science Modeling I (Spring 2023)*
*Three weeks into this semester, the instructor of this course had to take leave. In addition to my regular TA duties, I was put in charge of all the course logistics with regard to managing assignments, the course website, student questions, exams, and grades for 150 students.
**Outstanding TA runner-up in 2022 and Winner in 2023

SUPERVISION

Master's: Aleksander Markiewicz (University of Oxford 2025: *Operator Learning for Near Symmetric Matrices*)

INVITED TALKS

"o(n) Eigensolvers" September 2026 at ENLARGED in Didcot, UK.

"RandLinearAlgebra.jl: A cyberlaboratory for RandNLA" June 2026 at the RandNLA workshop in Lausanne, Switzerland.

"RandLinearAlgebra.jl" May 2026 at the Midwest Randomized Linear Algebra Workshop in Madison, WI.

"IterativeCUR: One Small Sketch for Big Matrix Approximations" November 2025 at the Applied and Computational Analysis (ACA) seminar in Cambridge, UK.

"Statistical Approaches to Numerical Linear Algebra" March 2025 at Optimization and Numerical Linear Algebra Exposé at the Rutherford Appleton Laboratory in Didcot, UK.

"Towards practical large-scale least squares solvers with Iterative Right Random Sketching" February 2023 at the Argonne National Labs LANS seminar in Chicago, Illinois (Virtual).

TALKS

"IterativeCUR: One Small Sketch for Big Matrix Approximations" May 2026 at the International Linear Algebra Society in Blacksburg, Virginia.

"IterativeCUR: One Small Sketch for Big Matrix Approximations" February 2026 at the Numerical Analysis Internal Seminar in Oxford, UK.

"Small Sketches for Big Matrix Approximations" September 2025 at ENUMATH in Heidelberg, Germany.

"Small Sketches for Big Matrix Approximations" July 2025 at SIAM Annual Meeting in Montreal, Canada.

"Small Sketches for Big Matrix Approximations" June 2025 at the 30th Biennial Numerical Analysis Conference in Glasgow, Scotland.

"Who needs a residual when an approximation will do?" December 2024 at the Numerical Analysis Internal Seminar in Oxford, UK.

"Tracking Progress in a Big Data World" October 2024 at SIAM Mathematics of Data Science in Atlanta, Georgia.

"Iterative Right Random Sketching for Large-Scale Logistic Regression" August 2024 at the Joint Statistical Meetings in Portland, Oregon.

"Computationally Efficient Tracking for Iterative Random Sketching" May 2024

at the SIAM Applied Linear Algebra Conference in Paris, France.

"Computationally Efficient Tracking for Iterative Random Sketching" April 2024 at the IFDS Ideas Forum in Madison, Wisconsin.

"Large-scale randomized iterative least squares" March 2023 at SIAM CSE in Amsterdam, Netherlands.

"Residual Tracking and Stopping for Iterative Random Sketching" April 2022 at Copper Mountain Conference on Iterative Methods in Copper Mountain, Colorado (Virtual).

HONORS & AWARDS

SIAM Student Travel Award (SIAM LA 2024) May 2024.

Outstanding TA May 2023.

Student Research Grant Competition UW - Madison February 2023.

Outstanding TA (Honorable Mention) May 2022.

SERVICE AND OUTREACH

Organizer: Numerical Analysis Group Internal Seminar (2025 - 2026), North-South Colloquium (2025 - 2026)

Committee Service: TA Training Redesign (2020), Space Committee (2021), Awards and Outreach Committee (2022), Postdoc representative for Health and Safety Committee (2025-2026), Postdoc head of Fridays at 4 [A department wide Ph.D./Postdoc focused seminar series] (2025-2026)

Statistics Graduate Student Association: Outreach Chair (Spring 2019 - Fall 2020), President (Fall 2020 - Spring 2023)

Journal Reviewer: IMA Journal of Numerical Analysis, SIAM Journal of Scientific Computing, IEEE Signal Processing Letters

Outreach: Wisconsin Science Festival (2020, 2022) (*Normal Distribution Learning Experience*: Built a giant four foot by four foot Galton Board and had kids drop ping-pong balls on it to learn the Central Limit Theorem. Then explained to kids how we could use this distribution to predict their heights.)

SKILLS

Computer Languages: Julia, R, C, C++, Python

APIs: CUDA, MPI, OpenMP, PETSc, PyTorch

SOFTWARE

RandLinearAlgebra (Lead Developer): A Julia Package for prototyping and benchmarking Randomized Linear Algebra algorithms. Funded by NSF grants 2309445 and 2309446 (PI: V. Patel) .

SOCIETY MEMBERSHIP	SIAM, American Statistical Association, International Linear Algebra Society, Phi Beta Kappa, Omicron Delta Epsilon